

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	29/09/2026
Team ID	SWTID-2026-4654
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	Create project structure, install dependencies, configure .env and MongoDB connection	3	High	
Sprint-1	Registration	USN-2	Register with name, email and password (bcryptjs hashing)	2	High	
Sprint-1	Login	USN-3	Login and issue JWT access and refresh tokens	2	High	
Sprint-1	Access Control	USN-4	Auth middleware and admin-only role check	3	High	
Sprint-2	Study Materials	USN-5	Upload material with multer (.txt, .md, .pdf, 5 MB)	5	High	
Sprint-2	Study Materials	USN-6	View and manage own materials	3	Medium	
Sprint-2	AI Summary	USN-7	Gemini integration and summary generation	5	High	
Sprint-3	AI Flashcards	USN-8	Generate and save flashcards	3	High	
Sprint-3	AI Quiz	USN-9	Generate and save multiple-choice quiz	5	High	
Sprint-3	AI Study Plan	USN-10	Generate study plan from goal, hours and days	5	High	
Sprint-4	History	USN-11	List previously generated resources	3	Medium	
Sprint-4	Administration	USN-12	Admin user management and usage monitoring	5	Medium	
Sprint-4	Quality	USN-13	Global error handling, validation and health check	3	Medium	
Sprint-4	Testing	USN-14	Postman API testing and documentation	3	Medium	

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	10	6 Days				
Sprint-2	13	6 Days				

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-3	13	6 Days				
Sprint-4	14	6 Days				

Sprint dates and completed points are left blank for the team to fill as sprints run.

Velocity:

Total planned story points are 50 across 4 sprints, an average of 12.5 points per sprint. With 6-day sprints (24 days), the average velocity is about 2.1 story points per day ($50 / 24$). Recalculate after each sprint using the points actually completed.

Burndown Chart:

A burndown chart plots remaining work against time. The ideal line falls evenly from the sprint's total story points to zero at the planned end date; the actual line is plotted from the points completed each day. Update it daily from the tracker above.